Operating Systems Project

Project Overview

In this project, students will work in teams of 5 members to implement and analyze fundamental Operating Systems algorithms. The project focuses on three main areas: CPU Scheduling Memory Allocation Page Replacement (Virtual Memory), Each team will collaboratively implement algorithms, take dynamic input from the user, and analyze performance.

Project Requirements

1) Algorithm Selection (Per Team) Each team must implement:
   - CPU Scheduling (choose ONE):

     FCFS (FIFO)

     SJF (Preemptive or Non-Preemptive)

     Round Robin

     Priority Scheduling
   - Memory Allocation (choose ONE):

     First Fit

     Best Fit

     Worst Fit
   - Page Replacement (choose ONE):

     FIFO

     LRU

     LFU

     MFU

     Clock
2) Input Requirements

   All inputs must be taken dynamically from the user, No hardcoded values allowed

   Examples:

   CPU Scheduling:

   Number of processes, Arrival time, Burst time, Priority (if needed), Time quantum

   Memory Allocation: Memory blocks (number & size), Processes (number & size)

   Page Replacement: Page reference string, Number of frames
3) Output Requirements

   Each algorithm must produce: Step-by-step execution Final results

   Examples:

   CPU Scheduling: Gantt Chart (text or visual), Waiting Time, Turnaround Time, Average Waiting & Turnaround Time

Memory Allocation: Allocation result, Internal/External Fragmentation

Page Replacement: Frame status per step, Total page faults

4) Visualization: Teams are encouraged to include: Gantt Charts, Tables, Graphs (e.g., using matplotlib)

5) Comparison & Analysis

Each team must compare: Their chosen algorithm with at least one alternative

Comparison must include: Performance differences, Advantages & disadvantages, Use cases

6) Report Submission (Team-Based) Each team will submit ONE report including:

Explanation of all implemented algorithms

Sample inputs & outputs

Comparison & analysis

Conclusion

❖ **Important Notes**

Project with python

No hardcoded inputs

All members must contribute

All team members must understand the full project, not only their part.